COMPUTER NETWORK: Data Link Layer

ASSIGNMENT 2

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Outline

1 Multiple Choice Questions (MCQs)

MCQ Q1: Network Device Layer Matching – Question

Question MCQ Q1

Question: Which of the following statements is NOT correct? **Options:**

(a) Every network interface card in Ethernet comes with a MAC address that is assured to be unique worldwide.

(b) A basic parity check mechanism is capable of identifying any odd count of bit errors in a transmitted code word.

(c) It is possible that a station in an Ethernet network wanting to transmit may never gain access to the medium.

(d) All of the above statements are correct

MCQ Q1: Network Device Layer Matching – Strategy

Problem-Solving Strategy

- Examine each claim against established networking principles.
- Ethernet MAC addresses are allocated in blocks to manufacturers - duplicates can occur, though infrequently.
- A single parity bit can indeed detect any odd number of bit errors.
- Ethernet employs CSMA/CD which under heavy load could theoretically lead to indefinite deferral.

MCQ Q1: Network Device Layer Matching – Solution

Solution

Answer: (a)

Option (a) is incorrect because Ethernet physical addresses are not guaranteed to be globally unique - they are assigned in blocks to manufacturers and there is a small possibility of duplicates. Options (b) and (c) are accurate statements.

MCQ Q2: Protocol Terminology – Question

Question MCQ Q2

Question: Identify the statement that is NOT accurate: **Options:**

(a) A dependable data delivery method may transmit several data units before receiving confirmations, rather than using a send-wait approach. This method is known as "Pipelining".

(b) An application exchanges information with the network through a programming interface called a "Socket".

(c) Since a web server retains no details about its visitors, such a server is described as "Stateful".

(d) The "Traceroute" utility helps find the number of intermediate nodes to a destination and the travel time for each hop.

MCQ Q2: Protocol Terminology – Strategy

Problem-Solving Strategy

- Pipelining: correctly describes sending multiple packets before acknowledgments.
- Socket: accurately defined as interface between process and network.
- HTTP/web server maintains no client information $\rightarrow$ this is "Stateless", not "Stateful".
- Traceroute: correctly identifies hops and round-trip times.

MCQ Q2: Protocol Terminology – Solution

Solution

Answer: (c)

A web server that keeps no record of client interactions is called "Stateless". The term "Stateful" would mean it maintains client information. Options (a), (b), and (d) are accurate descriptions.

MCQ Q3: Send-Wait Protocol Efficiency – Question

Question MCQ Q3

Question: Calculate the channel usage percentage for a send-and-wait protocol given: connection speed $= 2 \times 10^3$ bits per second, round trip time $= 40$ seconds, and each data unit $= 1000$ bytes. **Options:**

(a) 0%

(b) 5%

(c) 10%

(d) 15%

MCQ Q3: Send-Wait Protocol Efficiency – Strategy

Problem-Solving Strategy

- In send-and-wait protocol, only one data unit is transmitted per round trip.
- Effective throughput = Data unit size (in bits) $\div$ Round trip time.
- Channel usage = (Throughput $\div$ Link speed) $\times$ 100%.
- Remember to convert data unit from bytes to bits (multiply by 8).

MCQ Q3: Send-Wait Protocol Efficiency – Solution

Solution

Answer: (c) 10%

Data unit $= 1000$ bytes $= 1000 \times 8 = 8000$ bits

Throughput $= 8000$ bits $\div$ 40 sec $= 200$ bits/sec

Channel usage $= (200 \div 2000) \times 100 = 10\%$

MCQ Q4: Token Bucket Parameter – Question

Question MCQ Q4

Question: A traffic shaping mechanism uses a token container with peak rate $R = 20$ Mbps. The design requires that transmission at peak rate cannot exceed 5 continuous seconds, and total data transmitted in any 10-second interval cannot exceed 150 Mb. Determine the token generation rate (r) in Mbps. **Options:**

(a) 5

(b) 10

(c) 15

(d) 20

MCQ Q4: Token Bucket Parameter – Strategy

Problem-Solving Strategy

- At maximum rate R for 5 seconds: data transmitted $= R \times 5$.
- Within any 10-second window, total must be 150 Mb.
- The remaining 5 seconds can carry at most (150 - data from first 5 seconds).
- Token generation rate = remaining data $\div$ remaining time.

MCQ Q4: Token Bucket Parameter – Solution

Solution

Answer: (b) 10

At peak rate for 5 seconds: 20 Mbps $\times$ 5 sec $=$ 100 Mb

Remaining capacity in 10-sec window $=$ 150 Mb - 100 Mb $=$ 50 Mb

Token generation rate $=$ 50 Mb $\div$ 5 sec $=$ 10 Mbps

MCQ Q5: Window Size for Full Utilization – Question

Question MCQ Q5

Question: A communication channel spanning 1000 km operates at 10^9 bits/sec between two stations. Data units are fixed at 1250 bytes, and the sender continuously has data to transmit. No errors or losses occur. What window size achieves 100% efficiency in a sliding window protocol? Signal travels at 5 ms per kilometer. **Options:**

(a) 100

(b) 1000

(c) 110

(d) 1100

MCQ Q5: Window Size for Full Utilization – Strategy

Problem-Solving Strategy

- For 100% utilization: Window size $\times$ Transmission time $2 \times$ Propagation delay.
- Transmission time $=$ Data unit size (bits) $\div$ Data rate.
- Propagation delay $=$ Distance $\times$ Propagation speed per km.
- Window size $= (2 \times$ Propagation delay$) \div$ Transmission time.

MCQ Q5: Window Size for Full Utilization – Solution

Solution

Answer: (b) 1000

Data unit = 1250 bytes = 1250×8 = 10,000 bits

Transmission time = $10{,}000 \div 10$ = 10 sec = 10 μs

Propagation delay = 1000 km $\times$ 5 ms/km = 5000 ms = 5 sec

$2 \times$ Propagation delay = 10 sec

Required window size = 10 sec $\div$ 10 sec = 1,000,000?

Note: Based on answer key, answer is 1000 (likely unit inconsistency in problem).

MCQ Q6: CRC Computation – Question

Question MCQ Q6

Question: Using the data pattern "1010001101" and the generator polynomial $x^5 + x^4 + x^2 + 1$ for error detection, what is the complete transmitted message including the CRC? **Options:**

(a) 101000110101011

(b) 101000110101010

(c) 101000110101110

(d) 101000110111111

MCQ Q6: CRC Computation – Strategy

Problem-Solving Strategy

- Convert polynomial to binary: $x + x + x^2 + 1$ corresponds to 110101.
- Append 5 zeros (degree of polynomial) to the original message.
- Perform binary division (XOR operation) using the divisor.
- The remainder becomes the CRC, appended to the original message.

MCQ Q6: CRC Computation – Solution

Solution

Answer: (c) 101000110101110

Divisor: 110101

Message with zeros: 101000110100000

After binary division, remainder = 01110

Transmitted message = 1010001101 + 01110 = 101000110101110

MCQ Q7: Traffic Load Factor – Question

Question MCQ Q7

Question: A 10^7 bits/sec link extends 400 km and has buffer space for 2000 packets. Packets enter the queue at 4000 packets per second, with average packet size of 2000 bits. Determine the traffic intensity. **Options:**

(a) 0.2

(b) 0.4

(c) 0.6

(d) 0.8

MCQ Q7: Traffic Load Factor – Strategy

Problem-Solving Strategy

- Traffic intensity = Arrival rate (bits/sec) ÷ Link capacity (bits/sec).
- Arrival rate in bits/sec = Packets/sec × Bits per packet.
- Buffer size and distance are not needed for this calculation.

MCQ Q7: Traffic Load Factor – Solution

Solution

Answer: (d) 0.8

Arrival rate $= 4000$ packets/sec $\times$ 2000 bits/packet $= 8 \times 10$ bits/sec

Link capacity $= 10$ bits/sec

Traffic intensity $= (8 \times 10) \div (10) = 0.8$

MCQ Q8: Protocol Stack Overhead – Question

Question MCQ Q8

Question: In a network protocol stack with 10 layers, a 200 KB application message is sent. If headers consume 20% of the network bandwidth, what is the size of each layer's header in KB? **Options:**

(a) 5

(b) 6

(c) 10

(d) 15

MCQ Q8: Protocol Stack Overhead – Strategy

Problem-Solving Strategy

- Let header size per layer = H KB.
- Total header overhead = $10 \times$ H KB.
- Total transmitted data = Original message + Headers = $200 + 10H$ KB.
- Header fraction = $(10H) \div (200 + 10H) = 0.2$.
- Solve the equation for H.

MCQ Q8: Protocol Stack Overhead – Solution

Solution

Answer: (a) 5 KB

$10H \div (200 + 10H) = 1/5$

Cross multiply: $50H = 200 + 10H$

$40H = 200$

$H = 5$ KB

MCQ Q9: Long-Distance Link Utilization – Question

Question MCQ Q9

Question: A 1000-mile link operates at 1 Mbps capacity with no errors. A single sender continuously transmits 1250-byte frames. Signal propagation is 5 μs per mile. Calculate the channel utilization with a sliding window protocol using window size 7. **Options:**

(a) 25%

(b) 50%

(c) 75%

(d) 100%

MCQ Q9: Long-Distance Link Utilization – Strategy

Problem-Solving Strategy

- Compare total window transmission time with round trip time.
- If window transmission time $(2\times\text{propagation delay} + \text{transmission time})$, utilization $= 100\%$.
- Otherwise, utilization $=$ window transmission time $\div$ $(2\times\text{propagation} + \text{transmission time})$.
- Transmission time per frame $=$ frame size $\div$ data rate.

MCQ Q9: Long-Distance Link Utilization – Solution

Solution

Answer: (d) 100%

Frame size = 1250 bytes = 1250 × 8 = 10,000 bits

Transmission time per frame = 10,000 ÷ 10 = 0.01 sec = 10 ms

Window transmission time = 7 × 10 ms = 70 ms

Propagation delay = 1000 miles × 5 μs/mile = 5000 μs = 5 ms

Round trip propagation = 10 ms

70 ms ¿ (10 ms + 10 ms) = 20 ms, therefore utilization = 100%

MCQ Q10: Sliding Window Efficiency – Question

Question MCQ Q10

Question: Packet transmission requires 1 ms, while one-way propagation delay is 20 ms between endpoints. With a sliding window size of 4, what is the channel utilization? **Options:**

(a) 0.190

(b) 0.024

(c) 0.019

(d) 0.097

MCQ Q10: Sliding Window Efficiency – Strategy

Problem-Solving Strategy

- Channel utilization formula for sliding window protocols:
- Utilization = Window size ÷ (1 + 2 × Propagation delay ÷ Transmission time).
- Window size (N) = 4.
- Propagation delay (t_p) = $20ms$.
- Transmission time (t_t) = $1ms$.

MCQ Q10: Sliding Window Efficiency – Solution

Solution

Answer: (d) 0.097

Utilization $= 4 \div (1 + 2 \times 20 \div 1) = 4 \div (1 + 40) = 4 \div 41 = 0.097$